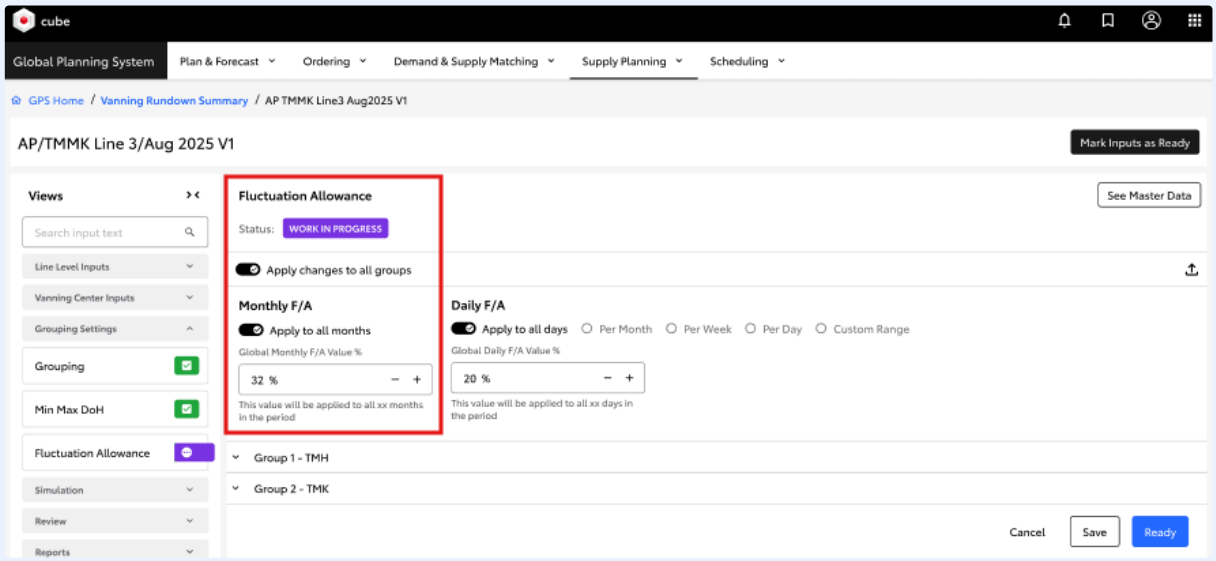
SPPL-262: Fluctuation Allowance Post API (Apply To All Groups False & Months True Only) technical workflow

Jira Ticket: SPPL-262: BE | Analysis | Update technical workflow for Monthly FA Post (Overall functionality) API including DB details DONE

This page provides comprehensive information about the Post API for Monthly FA on the Input params page, to view Monthly FA details for the scenario by groups for all months.



Supply Planning POST Monthly Fluctuation Allowance API Details:

1. **Hosted URL**: https://api.spln.cube<env>.toyota.com/<env>
2. **API Name:** <HOSTED_URL>/v/1/1/sp/monthly-fluctuation-allowance
3. **Path Param**: N/A
4. **Query Param**: N/A
5. **Method:** POST
6. **Authorization**: Bearer token (**required**)
7. **Lambda Name**: spln-<env>-update-monthly-fluctuation-allowance-v1-1
8. **API Request Body**:

```
{
  "scenarioId": "uniqueScenarioId",
  "userEmail": "user@toyota.com",
  "applyTo": "BY_GROUPS",
  "mode": "ALL_MONTHS",
  "data": [
    {
      "groupId": "group1uuid",
      "fa": { "value": 10 }
    },
    {
      "groupId": "group2uuid",
```

```
      "fa": { "value": 20 }
    }
  ]
}
```

9. **API Response:**
   - **Success response with status 200:**

```
{
  "message": "Successfully updated data."
}
```

   - **Error Response:**
     - **any internal server error occurred with status 500:**

```
{
  "errorMessage": "Internal Server Error"
}
```

     - **any unauthorized error occurred with status 401 (Managed by the AWS API Gateway integrated Custom Authorizer):**

```
{
  "message": "Unauthorized"
}
```

     - **If request is in-valid with status 400:**

```
{
  "errorMessage": [<"ValidationError: validation error message”>]
}
```

     - **List of possible validation error message:**
       1. "ValidationError: Request body cannot be empty.”
       2. "ValidationError: scenarioId is required and must be a uuid."
       3. "ValidationError: userEmail is required and must be a string."
       4. "ValidationError: Invalid userEmail."
       5. "ValidationError: Scenario doesn't exist."
       6. "ValidationError: applyTo is required and must be either ALL_GROUPS or BY_GROUPS.”
       7. "ValidationError: mode is required and must be either ALL_MONTHS or BY_MONTHS."
       8. "ValidationError: data is required and must be an array with atleast 1 and atmost 15 of objects."
       9. "ValidationError: groupId is required and must be a uuid."
       10. "ValidationError: fa is required and must be an object.”
       11. "ValidationError: value is required and must be a number between 0 to 100."

10. **Table:**
    a. **scenarios**
    b. **grouping**
    c. **group_scenario_mapper**
    d. **monthly_fa**
    e. **scenario_step_status**

SP Update Monthly Fluctuation Allowance Data API Workflow:

- **Technical workflow:**
  - Description:
    i. The Supply Planning user logs in to the application.
    ii. After the user provides creates a scenario, opens it and navigates to Fluctuation Allowance, a request is sent to the SP Post Monthly Fluctuation Allowance endpoint: ***sp/monthly-fluctuation-allowance*** with below payload:

```
{
  "scenarioId": "uniqueScenarioId",
  "userEmail": "user@toyota.com",
  "applyTo": "BY_GROUPS",
  "mode": "ALL_MONTHS",
  "data": [
    {
      "groupId": "group1uuid",
      "fa": { "value": 10 }
    },
    {
      "groupId": "group2uuid",
      "fa": { "value": 20 }
    }
  ]
}
```

    iii. After successful authentication, the **spln-update-*monthly-fluctuation-allowance*-v1-1** lambda is triggered by the SP update scenario Monthly Fluctuation Allowance endpoint: ***sp/monthly-fluctuation-allowance***. If authentication fails, an error with status 401 (*refer: error response-**any unauthorized error occurred with status 401**)* will be returned and sent to client.
    iv. After user authentication has been confirmed, the **spln-update-monthly-*fluctuation-allowance*-v1-1** lambda will perform validation on the request payload. If payload is valid will proceed to next step else throw validation error (*refer: error response**: Section if request is in-valid with status 400 and List of possible validation error message***). Below query is used to check if provided scenario is valid:

```
select * from supply_planning.scenarios where scenario_id = :scenarioId and
is_active = true;
```

v. After the request payload is successfully validated, the **spln-update-*fluctuation-allowance*-v1-1** lambda execute below logic:

1. As the input is for “ALL_MONTHS” & “BY_GROUPS”, get active groupIds for scenario (from mapper) using below query:

```
SELECT sgm.group_id AS "groupId"
    FROM supply_planning.group_scenario_mapper sgm
    JOIN supply_planning.grouping gp
      ON gp.group_id = sgm.group_id
    WHERE sgm.scenario_id = ${scenarioId}::uuid
      AND sgm.is_active = TRUE
      AND gp.is_active = TRUE
    ORDER BY sgm.group_id
```

2. Bulk UPDATE query with apply_to_all_months true — update monthly_fa table data by scenarioId, groupId using below query:

```
UPDATE TABLE supply_planning.monthly_fa SET
      monthly_fa_percent = ${value}::int,
      apply_to_all_months = false
      updated_by = ${userEmail}::text,
      last_updated_timestamp = CURRENT_TIMESTAMP
    WHERE scenario_id = :scenarioId
          AND group_id = :groupId;
```

3. Update the step & scenario status to “In Progress”. Only update scenario status to “In Progress” if its not already using scenarioData from query (iv) response.

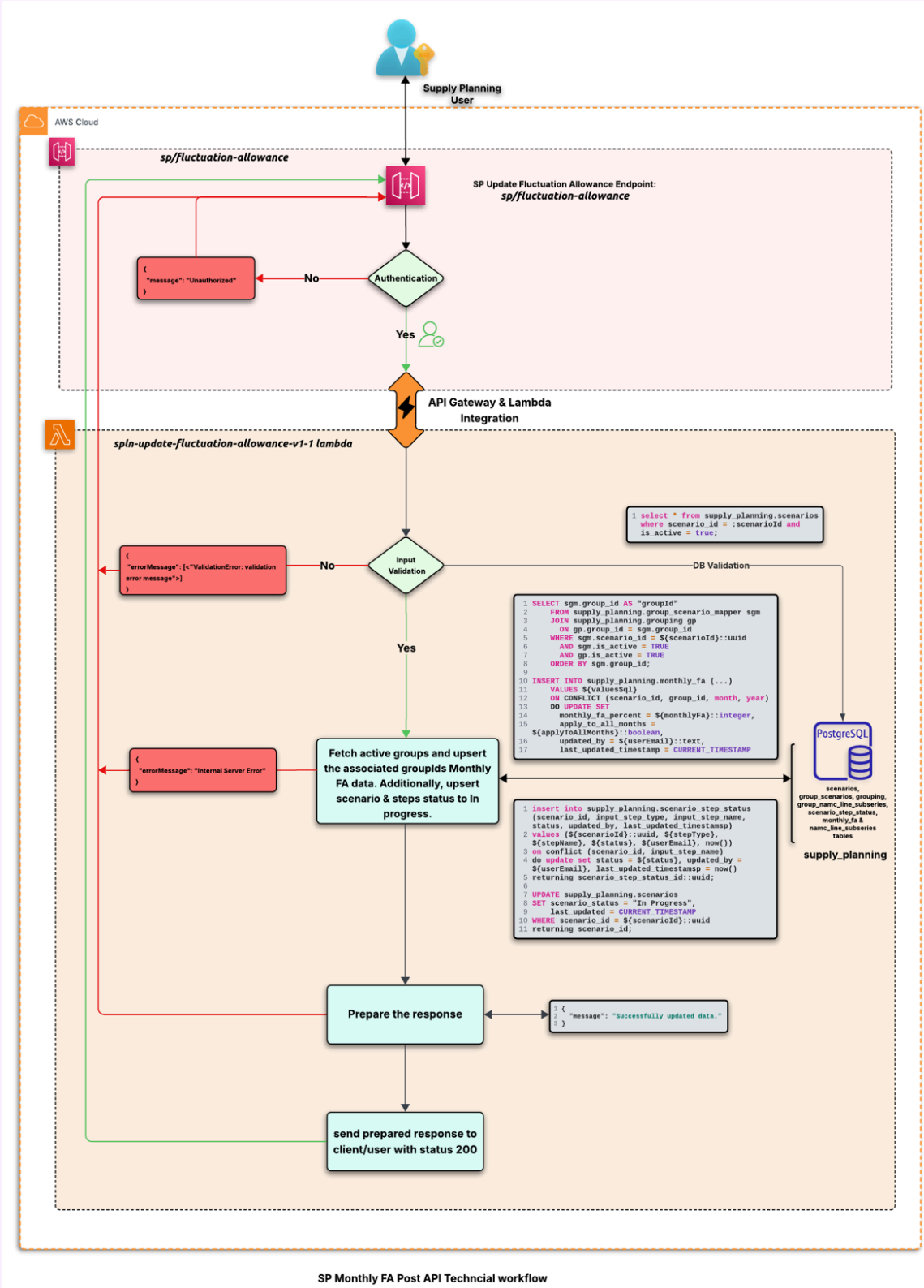
```
insert into supply_planning.scenario_step_status (scenario_id, input_step_type,
input_step_name, status, updated_by, last_updated_timestamsp)
values (${scenarioId}::uuid, ${stepType}, ${stepName}, ${status}, ${userEmail},
now())
on conflict (scenario_id, input_step_name)
do update set status = ${status}, updated_by = ${userEmail},
last_updated_timestamsp = now()
returning scenario_step_status_id::uuid;

UPDATE supply_planning.scenarios
SET scenario_status = "In Progress",
    last_updated = CURRENT_TIMESTAMP
WHERE scenario_id = ${scenarioId}::uuid
returning scenario_id;
```

vi. After updating the Monthly FA & status data, return the below response to the client with a status code of 200:

```
{
  "message": "Successfully updated data."
}
```

vii. If there is an error while fetching the data, an “Internal Server Error” with a status code of 500(*refer: error response-**any internal server error occurred with status***

500) will be returned. Otherwise, a successful response with a status code of 200 will be sent.



SP Monthly FA Post API Techncial workflow

Note:

We will be utilizing this API to update both file data and previous rundown data.

DB constraints:

```
CREATE INDEX IF NOT EXISTS idx_group_scenario_mapper_scenario_active
  ON supply_planning.group_scenario_mapper (scenario_id, is_active, group_id);

CREATE INDEX IF NOT EXISTS idx_monthy_fa_scenario_group_fa_month
  ON supply_planning.monthly_fa(scenario_id, group_id, fa_month);
```

Queries: